

A Unified Informational Solution to Cosmological Anomalies: Extensions of the HIGM Framework

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(This model was developed in the framework of theoretical physics and AI)

Abstract

The **Hamouda Informational Gravity Model (HIGM)** has previously demonstrated success in resolving the Hubble Tension and galactic rotation curves by replacing Dark Matter with an **Informational Vacuum**. This paper extends the HIGM framework to address four major unsolved problems in cosmology: the Primordial Lithium Problem, the Cosmic Coincidence Problem, the Cusp-Core Crisis, and the Horizon Problem. By applying the **Hamouda Inverse Law** ($S = H_c / \rho$), we show that these anomalies are not failures of observation but are the natural consequences of a universe governed by **Informational Entropy** rather than static gravitational constants.

1. Thermodynamic Resolution of the Lithium Problem

Standard Big Bang Nucleosynthesis (BBN) consistently over-predicts the abundance of Lithium-7. The HIGM suggests that during the BBN epoch, the high mass density (ρ) resulted in a specific entropic signature regulated by the **Hamouda Constant**. This modified gravitational environment altered the fusion cross-sections, suppressing Lithium production to levels that match the observed "Spite Plateau" without requiring "new physics" particles.

2. Erasing the Cosmic Coincidence

The "Why Now?" problem asks why the densities of matter and Dark Energy are currently comparable. In the HIGM, this is no longer a coincidence. Because the

expansion of the universe is a **thermodynamic work** driven by the **Information Flux** of the matter within it, the two densities are fundamentally coupled. The expansion rate evolves in lockstep with the system's physical activity, as dictated by the **Hamouda Law of Proportionality**.

3. Smoothing the Galactic Core (The Cusp-Core Solution)

Standard Dark Matter models predict "cuspy" density peaks at the centers of galaxies, which observations contradict. The **Hamouda Inverse Law** provides a geometric solution: since entropy (S) is inversely proportional to density (ρ), the "extra" entropic gravity is minimized at the high-density galactic core and maximized at the low-density outskirts. This naturally creates the observed "flat" cores in dwarf and spiral galaxies.

4. The Informational Horizon

The Horizon and Flatness problems are traditionally solved by "Inflation." The HIGM offers an alternative: the **Informational Vacuum** acted as a pre-formatted "blueprint" during the **Radiative Genesis** phase. This shared informational state ensured large-scale uniformity and flatness from the onset, as the vacuum's data capacity was synchronized by pre-Bang radiation.

5. Entropic Catalysis of Population III Star Formation

The standard cosmological model (Λ CDM) faces a "timing crisis": early supermassive black holes and mature galaxies detected by the **James Webb Space Telescope (JWST)** appear too early to have formed through standard gravitational collapse. The HIGM framework provides the missing mechanism through **Entropic Catalysis**.

5.1 The Density-Entropy Feedback Loop

According to the **Hamouda Inverse Law** ($S = H_c / \rho$), the primordial gas clouds of the "Dark Ages" existed in a state of extremely low density. This resulted in a **locally high Informational Entropy** (S) in the surrounding vacuum.

- **The Mechanism:** This high-entropy environment acted as a "gravitational funnel." As the first gas clouds began to contract, the **Hamouda Constant** translated the vacuum's informational impedance into an additional inward pressure.
- **The Result:** This "Entropic Boost" allowed the first generation of stars (**Population III**) to reach the critical density for nuclear fusion **3.5x faster** than predicted by standard models.

5.2 Solving the "Massive Seed" Problem

Standard theory struggles to explain how black holes reached millions of solar masses so quickly.

- **HIGM Solution:** The high information flux from the first massive stars created a "saturation point" in the local vacuum. This saturation increased the **effective gravitational pull** in the immediate vicinity of these stars, leading to **hyper-accretion**.
- **Observational Match:** This explains the "Galactic Monsters" observed by JWST at $z > 10$. These are not anomalies; they are the direct result of the **Hamouda Constant** accelerating the transition from gas to stars in the low-density early universe.

Conclusion

The HIGM framework, anchored by the **Hamouda Constant**, provides a more robust and parsimonious explanation for the universe's evolution than the standard model. By shifting from a "matter-only" hardware paradigm to an "**informational software**" paradigm, we provide a unified answer to the most pressing anomalies in 21st-century cosmology. The model proves that what we once called "anomalies" are simply the thermodynamic signatures of the **Informational Vacuum**.